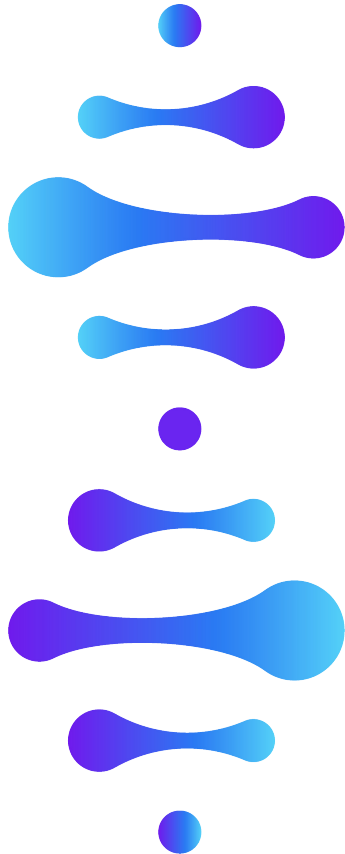




# IntellxxDNA™

Intelligent Science. Intelligent Living.

## Patient Report for:

Jenn Pei

DOB: 02/15/1994

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## Prepared by:

Nadia Taylor

Nadia Taylor MD Integrative Psychiatry

554 Clayton St

San Francisco, CA 94117

415-580-0558

**DISCLAIMER:** IntellxxDNA™ (IXXD) is a Clinical Decision Support (CDS) tool to be used solely by licensed health care providers. The function of the tool is to present your patient's genomic information, in the form of a report, alongside references to peer-reviewed clinical studies relating to specifically chosen genetic variants. It will also provide referenced information about potential strategies that may help you address these panels.

IntellxxDNA™ is NOT meant to be used to diagnose or treat illnesses. The reports are intended to be one factor of clinical decision-making and to be used along with a full history, physical exam and laboratory tests as appropriate. Providers must review patient information and make their own independent clinical decisions regarding treatment.

# Understanding your report

Your IntellxxDNA™ Report provides you and your clinician with the knowledge to help you achieve optimal health and wellness. This overview will help you understand the findings within your report.

## Genomics vs. Genetics:

Genetics is the study of individual genes and how they are inherited. Genetic diseases arise if you inherit 1 copy of a gene from each parent, like cystic fibrosis and sickle cell anemia. Most chronic diseases (such as diabetes, heart disease, cognitive decline, and even autism and ADHD) are not conveyed by any one gene but are genomic in origin. Genomics is the study of small changes in DNA that by themselves are not disease-causing but can contribute to disease as they interact with each other along with diet, lifestyle, and the environment.

**Alleles** are one of two or more versions of a specific genetic sequence on a chromosome. Individuals inherit one allele from each parent for a particular gene. The minor allele is the less common version but may convey either benefit or risk.

**Genes** are like a set of instructions that tells your body how to make specific things, like proteins. These proteins play important roles in how your body grows, functions, and behaves.

**Why this SNP is important** What makes it significant and how it can affect you.

|                     |                                |
|---------------------|--------------------------------|
| Gene Name           | COMT V158M                     |
| Minor Allele        | A                              |
| My Results          | AG                             |
| My Variants         | You have 1 minor allele        |
| Prevalence          | 50% of people have this result |
| Impact (Odds Ratio) | Medium<br>1.73                 |

### "Catechol-O-Methyltransferase" gene

*Your body does not break down the fight or flight molecules of adrenaline and norepinephrine as well. Too much of these stress molecules can increase the risk of a heart event.*

- This contributes to a decreased ability to break down adrenaline and dopamine. Variants have also been associated with increased risk of stress, but also improved memory.
- This variant has been associated with an increased risk for acute coronary syndrome, which is having a heart event while running or under stress.
- If you have elevated homocysteine and this variant, it can further contribute to risk.

*Supplements: Ashwagandha, rhodiola rosea, magnesium, zinc, EGCG, and genistein are supplements that may be beneficial and can be used with guidance on dosing, risks, and potential benefits.*

*Note: If you have two copies of this variant, you may want to avoid quercetin, as it can further decrease the breakdown of these fight or flight chemicals.*

**Prevalence** is how common this allelic combination is in the population.

**Impact** is the likelihood and degree to which the variant may contribute to genomic disease. Some SNPs have an Odds Ratio, while others do not.

**What you can do about it!** Your clinician will help you decide what is right for you.

**How this SNP works in your body.** Some of the science behind the SNP.

**Variant:** AKA "SNP" is a slight difference or alteration in a gene that can affect how a gene functions and can potentially impact an individual's traits or health. A SNP is a one "letter" or nucleotide change in our DNA. SNPs occur naturally in a population or can arise from mutations. Different variants have different functions, which can make a gene more or less active or affect the quality of the proteins made.

## Neurobehavioral Research

Autism has increased tremendously over the last few decades. In 1996, Autism was reported in approximately 3 out of every 1000 children in the US and 6 out of 1000 in Canada. In 2000, the CDC statistics showed the rates had increased to 1 out of 150 in the U.S. and the most recent CDC statistics place the risk of Autism in the 1 out of 36 to 1 out of 50 range. The reason for this tremendous increase in the prevalence of Autism is not due to any one component, but it is clear that environmental exposures, changes to our food supply, and changes to our microbiome all contribute heavily and that predisposition is conveyed by genomics, but the environment is an activating force.

There are over 1200 genes that have been associated with Autism. Many of them are very rare and for syndromes that are in the Autism spectrum<sup>278</sup>. These panels do not cover these rare genomic syndromes, but instead, the IntellxxDNA™ genomics panels cover the contributing genomic factors such as inflammation, detox, vitamin, methylation issues, and neurotransmitter issues. NOTE: This Neurobehavioral Research Panel is not a typical odds ratio panel. Many of the SNPs within this panel are still in more of a "preliminary" stage of research throughout the literature. Mechanisms, functions of variants, etc may not be completely elucidated, but this is still a culmination of the research to date.

This Autism research panel focuses on variants with studies showing Autism association but no studies yet meeting our stringent criteria of having odds ratios published showing more than 20% increase or decrease for the risk of Autism with studies being large enough to have significant p-values. In addition, this panel has variants being studied by Autism researchers that we are collaborating with around the world.

### Neurobehavioral Research Risk SNPs

|              |                                 |
|--------------|---------------------------------|
| Gene Name    | CYP1A2                          |
| Minor Allele | A                               |
| My Results   | AA                              |
| My Variants  | You have 2 minor alleles        |
| Prevalence   | 0.5% of people have this result |
| Impact       | High                            |

#### "Cytochrome P450 Family 1 Subfamily A Member 2" gene

**Key Point:** *Individuals with this SNP have decreased activity of CYP1A2, and therefore, slower metabolism. However, due to the combination of another SNP in the melatonin pathway, this SNP still leads to lower melatonin levels.*

- This variant increases the risk of sleep disorders for people with autism.

**Supplements:** *Melatonin can be considered.*

|              |                                 |
|--------------|---------------------------------|
| Gene Name    | GABRB3                          |
| Minor Allele | C                               |
| My Results   | CC                              |
| My Variants  | You have 2 minor alleles        |
| Prevalence   | 2.6% of people have this result |
| Impact       | High                            |

### "Gamma-aminobutyric Acid Type A Receptor Beta3 Subunit" gene

**Key Point:** GABA signaling is important during development and is involved in social skills, learning, motor skills, and various other developmental processes. Lower GABA signaling can lead to an increased risk of developmental disorders.

- Reduced number of GABA receptors in the brain have been found in Autism Spectrum disorders and are thought to be a significant contributor.
- This particular SNP correlated to:
  - TC having lowest risk of developmental symptoms
  - **TT has greatest risk of developmental symptoms such as inability to smile in response to mother and decreased ability to grasp things by self.**
  - CC intermediate risk of developmental symptoms.

**Supplements:** Many supplements, including L-theanine, taurine, ashwagandha, magnesium, vitamin B6, zinc, L-glutamine, luteolin, apigenin, ginkgo biloba, panax quinquefolius (American ginseng) and probiotics like lactobacillus rhamnosus and bifidobacteria can help increase GABA receptor and signaling. These supplements can be considered with guidance on dosing, risks, and potential benefits.

**Medications:** Bumetanide and oxytocin may help address this pathway. You and your clinician can talk about the risks versus benefits of these medications.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | PPARGC1A                         |
| Minor Allele | C                                |
| My Results   | CC                               |
| My Variants  | You have 2 minor alleles         |
| Prevalence   | 17.6% of people have this result |
| Impact       | High                             |

### "PPARG Coactivator 1 Alpha" gene

**Key Point:** PPARGC1A is involved in the production of mitochondria, which makes ATP (a crucial energy source) Individuals with this SNP have less of this protein and, therefore, are less able to make mitochondria and keep them healthy.

- This variant decreases the protein activity, thus decreasing antioxidant enzymes and decreasing mitochondrial creation (biogenesis).
- Problems with mitochondria are associated with autism.
- In an animal model, a high fat and sugar diet decreases this gene expression even more.

**Diet:** A low inflammatory diet that is low in saturated fat and simple sugars can support this pathway. In addition, intermittent fasting and omega3's may help support your mitochondria.

**Lifestyle:** Cold exposure and aerobic exercise can help increase this gene's expression. Maintaining a regular sleep schedule may help reduce risk.

**Supplements:** Omega 3's, alpha-lipoic acid, CoQ10, pterostilbene, resveratrol, and benfotiamine are supplements that can help boost mitochondrial function.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | HMOX1 -413A>T                    |
| Minor Allele | T                                |
| My Results   | TT                               |
| My Variants  | You have 2 minor alleles         |
| Prevalence   | 19.1% of people have this result |
| Impact       | High                             |

### "Heme Oxygenase 1" gene

**Key Point:** Individuals with this SNP have an overactive HMOX pathway. This overactive pathway leads to levels of iron and carbon monoxide that can be toxic to your brain. It can also increase what is known as oxidative stress (buildup of toxic oxygen molecules).

- This variant increases HMOX expression which can lead to an increased buildup of iron and carbon monoxide.

**Diet:** Incorporating garlic into your diet may be beneficial.

**Lifestyle:** Hyperbaric oxygen can help decrease carbon monoxide (which may build up with this variant) but should be considered carefully to determine the risk versus benefit.

**Supplements:** Lactoferrin is a supplement that can bind free iron. Supporting your mitochondria with sulforaphane, glutathione, SOD, and hydrogen water may also be beneficial and can be considered with guidance on dosing.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | MMP9                             |
| Minor Allele | T                                |
| My Results   | CT                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 22.4% of people have this result |
| Impact       | Medium                           |

You have a genomic variant(s) in this "Matrix Metalloproteinase 9" pathway.

**Key Point:** This gene variant leads to higher levels of Matrix Metalloproteinase 9 (MMP9). This can lead to brain inflammation, and high levels of MMP9 have been seen in people with Autism Spectrum Disorder.

- The T gene variant is associated with higher levels of MMP9
- MMP9 has many effects, including disrupting the protective blood-brain barrier and increasing brain inflammation, like TNF -α and IL-1β.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | GRIA3                            |
| Minor Allele | C                                |
| My Results   | C^                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 31.7% of people have this result |
| Impact       | Medium                           |

### "Glutamate Ionotropic Receptor AMPA Type Subunit 3" gene

**Key Point:** Individuals with this SNP have less of this glutamate (excitatory brain chemical) receptor. This receptor is responsible for keeping the balance between GABA (calm brain chemical) and glutamate signaling. Difficulties with this balance can lead to high levels of glutamate and low levels of GABA which can be irritating/toxic to the brain.

- Normal stimulation of these receptors promote wakefulness, while over-activation has been implicated in damage to the brain, relating to traumatic brain injuries, seizures and memory issues.
- This gene pathway has been associated with autism and sleep.

**Diet:** Monosodium glutamate (MSG) may increase glutamate levels so limiting the consumption of it may be beneficial. MSG is a common flavor additive.

**Supplements:** Ways to increase GABA can include l-theanine, taurine, magnesium, vitamin B6 and the probiotic lactobacillus rhamnosus. NAC and lithium can help lower glutamate. These can be considered with guidance on dosing and risks versus benefits.

**Medications:** There are medications that can help decrease glutamate, including Amantadine, and Memantine. This can be discussed with your clinician.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | PPARGC1A                         |
| Minor Allele | T                                |
| My Results   | CT                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 43.6% of people have this result |
| Impact       | Medium                           |

### "PPARG Coactivator 1 Alpha" gene

**Key Point:** PPARGC1A is involved in the production of mitochondria, which makes ATP (a crucial energy source). Individuals with this SNP have less of this protein and, therefore, are less able to make mitochondria or keep their mitochondria healthy.

- This variant decreases the protein activity, thus decreasing antioxidant enzymes and decreasing mitochondrial creation (biogenesis).
- Problems with mitochondria are associated with autism.
- In an animal model, a high fat and sugar diet decreases this gene expression even more.

**Diet:** A low inflammatory diet that is low in saturated fat and simple sugars can support this pathway. In addition, intermittent fasting and omega3's may help support your mitochondria.

**Lifestyle:** Cold exposure and aerobic exercise can help increase this gene's expression.

**Supplements:** Omega 3's, alpha-lipoic acid, CoQ10, pterostilbene, resveratrol, and benfotiamine are supplements that can help boost mitochondrial function.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | CNTNAP2                          |
| Minor Allele | T                                |
| My Results   | AT                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 46.6% of people have this result |
| Impact       | Medium                           |

#### "Contactin Associated Protein Like 2" gene

**Key Point:** Individuals with this SNP have an increased risk of language delays and attention deficit disorder due to the lowered ability for nerves to communicate with each other. This also may impair GABA (calm brain chemical) signaling, which can further increase the risk of autism symptoms.

- This variant is associated with speech and language delays in autism.
- This variant may also lead to less GABA (calm brain chemical) release.

**Supplements:** Retinoic acid (Vitamin A) and vitamin D can help push this pathway. In addition, the probiotic *Lactobacillus reuteri* can help increase oxytocin which this variant decreases. Increasing GABA signaling with supplements like L-theanine may also be of benefit. Note: High levels of TNA alpha decrease nerve growth factors so your clinician may consider incorporating treatments that lower this (magnesium threonate, etc.).  
**Medications:** Oxytocin may be considered with guidance on risks versus benefits.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | GABRG3                           |
| Minor Allele | T                                |
| My Results   | AT                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 48.4% of people have this result |
| Impact       | Medium                           |

#### "Gamma-aminobutyric Acid Type A Receptor Gamma3 Subunit" gene

**Key Point:** GABA signaling is important during development and is involved in social skills, learning, motor skills, and various other developmental processes. This variant may decrease GABA signaling. Lower GABA signaling can affect these processes and lead to an increased risk of developmental disorders.

- GABA receptor subunit. Even having one copy of T gene is associated with worse imitative behavior.
- Not 100% clear from research if having this variant means that the "GABA pathway" activity is low but given this gene's association with autism, may be reasonable to increase GABA in these children.

**Supplements:** Many supplements, including L-theanine, taurine, ashwagandha, magnesium, vitamin B6, zinc, L-glutamine, luteolin, apigenin, ginkgo biloba, panax quinquefolius (American ginseng) and probiotics like *Lactobacillus rhamnosus* and *Bifidobacteria* can help increase GABA receptor and signaling. These supplements can be considered with guidance on dosing, risks, and potential benefits.  
**Medications:** Bumetanide and oxytocin may help address this pathway. You and your clinician can talk about the risks versus benefits of these medications.

|              |                                  |
|--------------|----------------------------------|
| Gene Name    | GABRA5                           |
| Minor Allele | C                                |
| My Results   | CT                               |
| My Variants  | You have 1 minor allele          |
| Prevalence   | 48.7% of people have this result |
| Impact       | Medium                           |

#### "Gamma-aminobutyric Acid Type A Receptor Alpha5 Subunit" gene

**Key Point:** GABA signaling is important during development and is involved in social skills, learning, motor skills, and various other developmental processes. This variant may decrease GABA signaling. Lower GABA signaling can affect these processes and lead to an increased risk of developmental disorders.

- GABA receptor subunit. Linked to verbal communication. Interestingly, best verbal communication was in CT variant. However CC and TT variants linked with decrease in verbal communication.

**Supplements:** Many supplements, including L-theanine, taurine, ashwagandha, magnesium, vitamin B6, zinc, L-glutamine, luteolin, apigenin, ginkgo biloba, panax quinquefolius (American ginseng) and probiotics like *Lactobacillus rhamnosus* and *Bifidobacteria* can help increase GABA receptor and signaling. These supplements can be considered with guidance on dosing, risks, and potential benefits.  
**Medications:** Bumetanide and oxytocin may help address this pathway. You and your clinician can talk about the risks versus benefits of these medications.

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| Gene Name    | ASTN2                                                                             |
| Minor Allele | G                                                                                 |
| My Results   | CG                                                                                |
| My Variants  | You have 1 minor allele                                                           |
| Prevalence   | 45.5% of people have this result                                                  |
| Impact       |  |

**"Astrolactin 2" gene**

**Key Point:** *ASTN2 is involved in the developmental processes of neurons. Individuals with this SNP may have a decreased risk of neurodevelopmental disorders, such as autism.*

- This variant is associated with a decreased risk of neurodevelopmental disorders, including autism.

## Autism Spectrum Contributing Factors

Autism has increased tremendously over the last few decades. While just 25 years ago Autism was reported in about 3 in 1000 children in the US, the recent CDC data place the risk of Autism to between 1 out of 36 and 1 out of 50 range. There are many components contributing to this increase, but it is clear that environmental exposures, changes to our food supply, and changes to our microbiome all contribute heavily. It is also important to consider that one's genomics also plays a role in relation to one's environment.

While there are quite a few genetic "syndromes" that "cause" autism, these single gene pathogenic mutations make up a very small percentage of autism. For the majority of children, autism is triggered by a whole range of different genetic variants interacting with environmental factors. The majority of pathways below that have a high risk in contributing to Autism relate either to glutathione pathways (master detox pathway but also is a regulator of inflammation in the brain), interleukin inflammatory regulators, glutamate (an irritating neurotransmitter) pathways, or other oxidative stress pathways. Hundreds of other SNPs and pathways including those relating to gut permeability, autoimmunity, food tolerances, vitamin levels, oxytocin, melatonin receptors etc are also highly important. Optimizing brain health in general by looking at BDNF, choline metabolism, other detox pathways also is important. These pathways are found in other panels of the report.

Because controlling inflammation prior to addressing detoxification and nutrient deficiencies is essential to good outcomes with Autism, the Autism panel is not in a straight "odds ratio" format, with the highest odds ratios at the top. Instead, this panel first presents inflammation and oxidative stress-related SNPs, followed by detoxification, and then related neurotransmitter and other nutrient related pathways needed to "rebuild" the healthy brain. NOTE: Zinc and Vitamin D and Vitamin D receptors (which regulate hundreds of other genes throughout the genome) as well as additional detox and nutrient related SNPs are also found in their own panels. This panel only contains the most well-studied variants that have published odds ratios.

The Autism genomics presented through our IntellxxDNA™ reports reflect the amazing work of Dr. Heather Way and her team of clinicians as well as the work of many other dedicated researchers throughout the world. Dr. Way has gotten hundreds of autistic children from non-verbal to verbal using genomics as the lynchpin. However, addressing genomics without addressing nutrition and the microbiome will only provide limited success, therefore working with a nutritionist experienced with Autism to improve eating habits is crucial for optimization of function with Autistic children. Please see special handouts for transitioning of diet in Autistic children as this needs to be done very slowly in order to not create withdrawal symptoms (from dairy and gluten) and to create optimal success. More diet discussion is found in our celiac/gluten/dairy pathways.

### Autism Spectrum Contributing Factors Risk SNPs

|                     |                                  |
|---------------------|----------------------------------|
| Gene Name           | GLRX                             |
| Minor Allele        | C                                |
| My Results          | CC                               |
| My Variants         | You have 2 minor alleles         |
| Prevalence          | 20.2% of people have this result |
| Impact (Odds Ratio) | High<br>3.78                     |

#### "Glutaredoxin" gene

***This variant leads to increased oxidative stress (buildup of "toxic" oxygen molecules), inflammation, and decreased glutathione (master detox molecule) recycling. This can lead to mitochondrial (the powerhouse of the cell) dysfunction. It is significant for Autism if combined with other variants in this panel.***

- This increased risk of Autism ONLY applies if 2 variants are present in GLRX3 rs3750834, OR if 2 variants are present in the CTH rs12737233 SNP's above.

**Supplements:** Vitamin C, hydrogen water, SOD, PQQ, sulforaphane, and glutathione are examples of supplements that can help detoxification and can be considered with guidance on dosing.

|                     |                                  |
|---------------------|----------------------------------|
| Gene Name           | IL1B                             |
| Minor Allele        | A                                |
| My Results          | AG                               |
| My Variants         | You have 1 minor allele          |
| Prevalence          | 45.4% of people have this result |
| Impact (Odds Ratio) | High<br>3.02                     |

#### "Interleukin 1 Beta" gene

**IL1B is an inflammatory molecule that activates inflammatory mediators called inflammasomes, which leads to increased inflammatory response. This variant is associated with increased IL1B which can contribute to autism.**

- This variant increases autism risk.
- Resolvins/SPMs, fish oil, sulforaphane, and blueberries/grapes/other foods with pterostilbene can help decrease IL1B.

**Supplements:** High-dose fish oil with EPA, sulforaphane, and resolvins/SPMs are examples of supplements that can help lower IL1B and can be considered with guidance on dosing.

|                     |                                  |
|---------------------|----------------------------------|
| Gene Name           | SGSM2                            |
| Minor Allele        | T                                |
| My Results          | TG                               |
| My Variants         | You have 1 minor allele          |
| Prevalence          | 49.3% of people have this result |
| Impact (Odds Ratio) | Medium<br>1.53                   |

#### "Small G Protein Signaling Modulator 2" gene

**You have a decreased ability to repair your mitochondria, which are crucial for providing energy to cells. Mitochondrial dysfunction has been associated with autism.**

- This variant is associated with an increased risk of autism.

**Supplements:** Helping support your mitochondria with supplements like CoQ10, ALA, acetyl L-carnitine, Benfotiamine, and NAC (if not SHANK3) may be beneficial with guidance on dosing.

|                     |                                  |
|---------------------|----------------------------------|
| Gene Name           | SLC25A12                         |
| Minor Allele        | T                                |
| My Results          | TC                               |
| My Variants         | You have 1 minor allele          |
| Prevalence          | 24.4% of people have this result |
| Impact (Odds Ratio) | Medium<br>1.21                   |

#### "Solute Carrier Family 25 Member 12" gene

**SLC25A12 helps carry the building blocks (amino acids) aspartate and glutamate into the mitochondria, which allows for the production of nicotinamide adenine dinucleotide (NAD). NAD is important for the removal of toxins and energy production. If you have two variants you have better ability to carry glutamate into your mitochondria and produce NAD. This can decrease your risk of autism.**

- Two variants are associated with a decreased risk of autism.

**Lifestyle if 1 (or 0) variant:** Hyperbaric oxygen therapy can be considered as a benefit for this pathway; however, consider the risks and benefits with your clinician.

**Supplements if 1 (or 0) variant:** NAD and L-glutamine can help support energy production if you have this variant. Oxaloacetate, citicoline, saffron, and hydrogen water can help support nerve myelination. Myelination is essential for your nerves to transmit information to each other.

## Autism Spectrum Contributing Factors Benefit SNPs

|                     |                                         |
|---------------------|-----------------------------------------|
| Gene Name           | MET                                     |
| Minor Allele        | G                                       |
| My Results          | GG                                      |
| My Variants         | You have <b>2</b> minor alleles         |
| Prevalence          | <b>19.7%</b> of people have this result |
| Impact (Odds Ratio) | <div>Helpful</div> <div>0.52</div>      |

### "MET proto-oncogene, receptor tyrosine kinase" gene

- Involved in development and migration of muscles and neural precursors during pregnancy and early childhood development.
- Promoter SNP. The major allele C is associated with increased relative risk of Autism.

|                     |                                       |
|---------------------|---------------------------------------|
| Gene Name           | GLRX3                                 |
| Minor Allele        | G                                     |
| My Results          | AG                                    |
| My Variants         | You have <b>1</b> minor allele        |
| Prevalence          | <b>24%</b> of people have this result |
| Impact (Odds Ratio) | <div>Helpful</div> <div>0.69</div>    |

### "Glutaredoxin 3 " gene

***GLRX3 is involved in inserting iron needed into different proteins and the mitochondria (powerhouse of the cell). You have increased ability to attach iron properly, which can decrease autism risk.***

- This is a benefit SNP - having 1 or 2 copies of the G allele is associated with decreased risk for autism.

# Folate Receptor Antibody and Cerebral Folate Deficiency

It is important to get enough folate (Vitamin B9) to your brain. Low folate increases the risk of things like autism spectrum disorder (ASD), depression, and cognitive decline. In addition, folate is essential for fetal brain development. Sometimes, someone may have enough folate in his or her blood, but have low amounts in the brain. This is called cerebral folate deficiency (CFD). CFD has been associated with folate receptor autoantibodies (FRAA's). These are antibodies that block the ability for folate to transport to the brain. These autoantibodies are associated with ASD.

While these antibodies block folate transport to the brain is blocked, folinic acid, and active form of folate can still cross the blood brain barrier to get to the brain. Because of this, folinic acid is a suitable option if you have FRAA's. This panel looks at small changes in DNA called SNP's or variants that can contribute to either an increased or decreased risk of you developing FRAA's. In the presence of these variants, during times like pregnancy it may be reasonable to take a prenatal vitamin that contains folinic acid.

## Folate Receptor Antibody and Cerebral Folate Deficiency Risk SNPs

|              |                                |
|--------------|--------------------------------|
| Gene Name    | MTHFD2                         |
| Minor Allele | G                              |
| My Results   | AG                             |
| My Variants  | You have 1 minor allele        |
| Prevalence   | 50% of people have this result |
| Impact       | Medium                         |

### "Methylenetetrahydrofolate Dehydrogenase/Cyclohydrolase 2" gene

***If you have two variants you may have an increased risk of developing folate receptor autoantibodies.***

***Supplements (if two variants): Folinic acid is the most direct intervention for this pathway if cerebral folate deficiency is suspected, and can be considered with guidance on dosing.***